

7.b) Draw the block diagram of a communication system & explain the function of each stage (06 Marks)

### Block Diagram

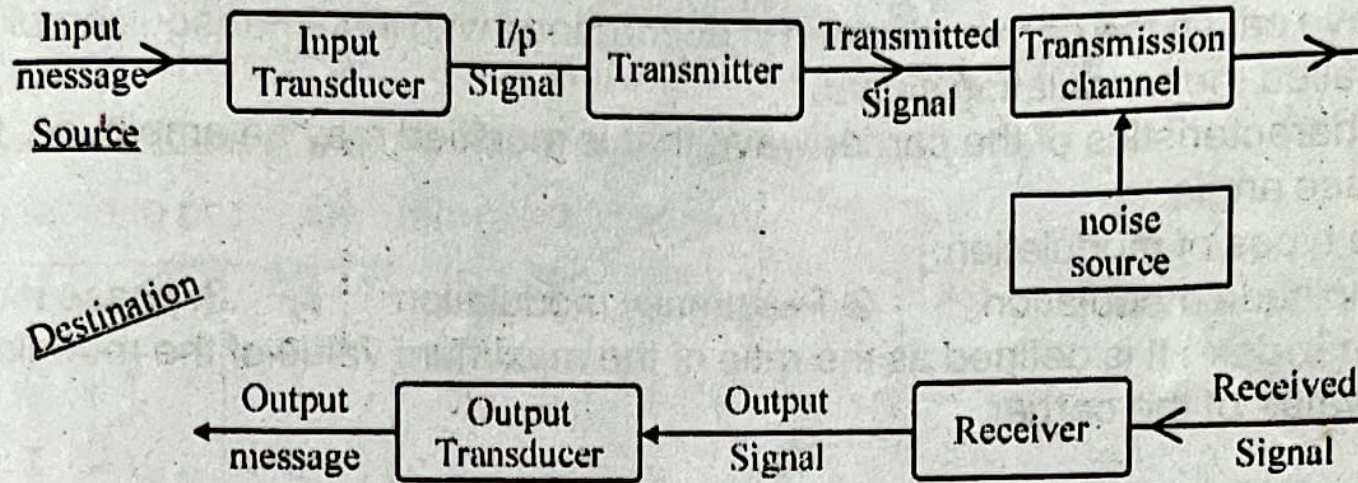


Fig 7(a)

The elements of communication system are as follows:

- Input transducer:** Is a device used to convert the physical parameters in to an electrical signal.
- Transmitter:** It processes the input signal in a form suitable for transmission over the communication channel. This operation is termed as modulation.
- Communication channel :** It provides physical connection between the transmitter and receiver.

There are 2 types of channels:

- Point to point channel:** Telephone wires, cables.
  - Broad cast channel:** Satellite in geostationary orbit.
- Noise :** Is an unwanted energy that gets added to the message signal during transmission over the channel It is random in nature & has its greatest effect when the message signal is weakest.
  - Receiver:** It processes the signal & produces an estimate of original message signal This operation is called demodulation.
  - o/p transducer:** Is a device used to convert electrical signal in to original physical parameter.

7.c) Distinguish between AM & FM.

**7.d) What is modulation? What are the three types of modulation & also Define modulation index? (04 Marks)**

**Sol: Modulation:** It is defined as the process by which some characteristic or property of a high frequency sine wave called the carrier is varied in accordance with instantaneous amplitude of the message signal called the modulating signal.

- \* The characteristics of the carrier wave that is modified may be amplitude, frequency or phase angle.
- \* Three types of modulation:
  1. Amplitude modulation
  2. Frequency modulation
  3. phase modulation

**Modulation index :** It is defined as the ratio of the maximum value of the modulating signal to the maximum value of the carrier.

$$m = \frac{V_m}{V_c}$$

**7.b) Explain the need for modulation****(08 Marks)**

Sol: Direct transmission of information signals such as voice, video or binary data would result in catastrophic interference problems when transmission distances are large.

- ❖ To carry out reliable long - distance radio communication. It is necessary to modulate the signal because of following reasons.

**1. To reduce the height of antenna:**

- ❖ For efficient radiation & reception the antenna's should have heights comparable to quarter wave length of the frequency used

$$\text{Minimum height of antenna} = \lambda/4 = C/4f$$

- ❖ Where  $\lambda$  is the wavelength,  $f$  is the frequency of the signal &  $c$  is velocity of electromagnetic wave (light).

**2. To avoid mixing of Signals.**

- ❖ We know that frequency range of audio signal is from 20 Hz to 20 KHz. If there are several stations operating over this same frequency range, the programme of different stations will get mixed up.

**3. To increase the range of communication:**

- ❖ At low frequencies radiation is poor & signal gets highly attenuated. Modulation makes it easier for signals to be transmitted over long distance.

**4. To allow multiplexing of signals:**

- ❖ Multiplexing means transmission of two or more signals simultaneously over the same channel.

**Ex:** Radio station broadcasting the signal in MW & SW band simultaneously.

**5. To allow adjustments in the BW (Band width):**

- ❖ BW of a modulating signal may be made smaller or larger than the original signal.

**6. Improve quality of reception:**

- ❖ Reduction in noise improves the quality of reception.

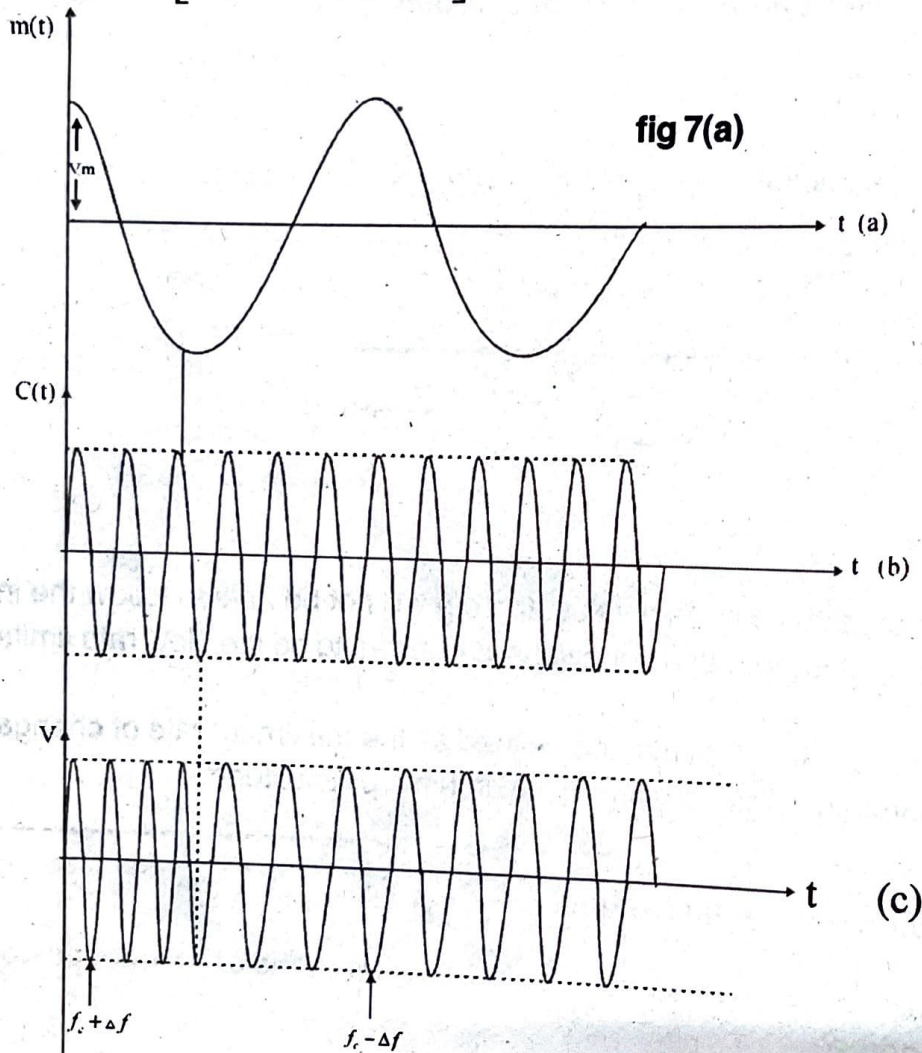
- 7.b) Explain with sketches, the concept of frequency modulation. From the fundamentals, derive an expression for the frequency Modulated voltage. What are its advantages over Amplitude modulation. **(10 Marks)**

Sol: Frequency modulation: The frequency of carrier is varied linearly with the message or information according to the relation.

$$f = f_c + k_f V_m$$

Where  $K_f$  = frequency sensitivity of modulator Instantaneous value of frequency modulated wave is given by

$$V = V_c \cos \left[ \omega_c t + 2\pi k_f \int_0^t V_m dt \right]$$



**7.b) Explain with neat wave form the principle of amplitude modulation. Write the expressions for Am wave. (06 Marks)**

**Sol:** The amplitude of a sinusoidal carrier wave is varied in accordance with the information bearing signal.

**Principle of Amplitude modulation:**

❖ In amplitude modulation the amplitude of the carrier wave is altered in accordance with the instantaneous amplitude of the modulating wave.

❖ Let the carrier voltage is given by

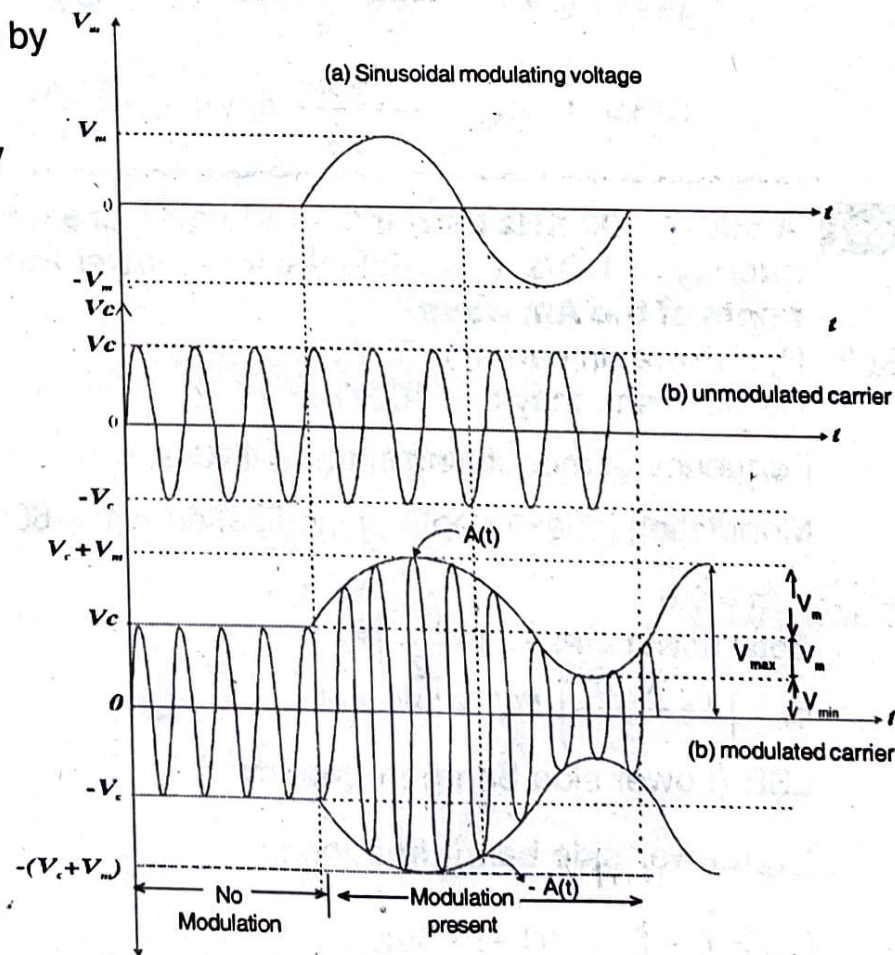
$$V_c = V_c \sin \omega_c t \text{ and}$$

Modulating voltage is given by

$$V_m = V_m \sin \omega_m t$$

$$V_m < V_c \text{ to avoid distortion}$$

**fig 7(a)**



- e. **Receiver:** It processes the signal & produces an estimate of original message signal This operation is called demodulation.
- f. **o/p transducer:** Is a device used to convert electrical signal in to original physical para meter.
- 7.c) **Distinguish between AM & FM.**

| AM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | FM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ol style="list-style-type: none"> <li>1. In Am, amplitude of carrier varies while its frequency remains constant.</li> <li>2. Modulation index (<math>m_a</math>) can have values from 0 to 1 only.</li> <li>3. Am has only 2 side bands</li> <li>4. Bandwidth is twice the highest modulating frequency. <math>B_w = 2 f_m</math></li> <li>5. Carrier frequency is lower RF 500 KHz to 3 MHz.</li> <li>6. Susceptible to noise</li> <li>7. Propagation is by ground waves &amp; sky waves.</li> <li>8. Am transmission covers long distances.</li> <li>9. Require less complex &amp; less expensive equipment.</li> <li>10. Area of reception is large.</li> </ol> | <ol style="list-style-type: none"> <li>1. In Fm, frequency of carries changes while its amplitude remains constant.</li> <li>2. Modulation index (<math>m_f</math>) is much greater than 1.</li> <li>3. FM has infinite number of side bands.</li> <li>4. Band width is much larger than Am<br/><math>B_w = 2(mf + f_m)</math></li> <li>5. carrier frequency is very high VHF &amp; UHF (88 to 100 MHz)</li> <li>6. More immune to noise</li> <li>7. Propagation is by space waves.</li> <li>8. FM transmission covers small distances only.</li> <li>9. Requires more complex &amp; expensive equipment.</li> <li>10. Area of reception in smaller than AM.</li> </ol> |